November 22, 2024

Cala Health, Inc.
% Amanda Johnston
Regulatory Counsel for Cala Health, Inc.
Gardner Law, PLLC
423 Main Street
Stillwater, Minnesota 55082

Re: K242259
Trade/Device Name: Cala kIQ
Regulation Number: 21 CFR 882.5897
Regulation Name: External upper limb tremor stimulator
Regulatory Class: Class II
Product Code: QBC
Dated: July 31, 2024
Received: July 31, 2024

Dear Amanda Johnston:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device"

(https://www.fda.gov/media/99812/download) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (https://www.fda.gov/media/99785/download).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance) and CDRH Learn (https://www.fda.gov/training-and-continuing-education/cdrh-learn). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

Jitendra V. Virani -S

CDR Jitendra Virani, MS, MBA
Assistant Director
DHT5B: Division of Neuromodulation and
Physical Medicine Devices
OHT5: Office of Neurological and
Physical Medicine Devices
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

DEPARTMENT OF HEALTH AND HUMAN SERVICES
Food and Drug Administration

Indications for Use

Form Approved: OMB No. 0910-0120
Expiration Date: 07/31/2026
See PRA Statement below.

Submission Number *(if known)*

K242259

Device Name

Cala kIQ

Indications for Use *(Describe)*

Cala kIQ is indicated to aid in the temporary relief of hand tremors in the treated hand following stimulation in adults with essential tremor.

Cala kIQ is indicated to aid in the temporary relief of postural and kinetic hand tremor symptoms that impact some activities of daily living in the treated hand following stimulation in adults with Parkinson's disease.

Type of Use *(Select one or both, as applicable)*

☒ Prescription Use (Part 21 CFR 801 Subpart D) ☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

This section applies only to requirements of the Paperwork Reduction Act of 1995.

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510(k) Summary

I. SUBMITTER

Manufacturer: Cala Health, Inc.
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Date Prepared: October 28, 2024

II. SUBJECT DEVICE

Trade Name: Cala kIQ™

Regulation Name: External upper limb tremor stimulator

Regulatory Class: Class II

Regulation Number: 21 CFR 882.5897

Product Code: QBC

III. PREDICATE DEVICE

Predicate Device: Cala kIQ™

Prior Submissions: K222237

Indications for Use: Cala kIQ is indicated to aid in the temporary relief of hand tremors in the treated hand following stimulation in adults with essential tremor.

Cala kIQ is indicated to aid in the temporary relief of postural and kinetic hand tremor symptoms that impact some activities of daily living in the treated hand following stimulation in adults with Parkinson's disease.

IV. DEVICE DESCRIPTION

Cala kIQ Gen 2 is a small, lightweight, wrist-worn stimulator device designed to aid in the temporary relief of hand tremors by applying a calibrated Transcutaneous Afferent Patterned Stimulation (TAPS) to the median and radial nerves of a patient's wrist.

The TAPS output pattern is calibrated to the individual patient's tremor frequency. During device setup, the patient performs a postural hold, during which the onboard sensors measure the tremor frequency, which is then used to customize the TAPS output pattern to deliver therapy to the median and radial nerves at the appropriate, patient-specific, optimal stimulation frequency.

Cala kIQ Gen 2 is comprised of the following components:

- Stimulator: Contains sensors, electronics, and firmware for delivering TAPS therapy and provides user interface
- Band: Contains multi-use, conductive electrodes spaced at intervals to align with targeted nerves and attachment points for the Cala kIQ Gen 2 Stimulator
- Base Station: Provides charging function and contains an indicator light to alert the user to operating conditions.

The Cala kIQ Gen 2 Stimulator contains the electronics, sensors, and firmware for calibrating and delivering TAPS therapy. The patient uses the Cala kIQ Gen 2 Stimulator buttons to complete device calibration, adjust the stimulation amplitude, and initiate therapy sessions. The Cala kIQ Gen 2 Stimulator is attached to the Cala kIQ Gen 2 Band, which includes embedded electrodes placed at appropriate intervals to properly target the median and radial nerves. To accommodate a broad distribution of wrist sizes, the band is available in three sizes (small, medium, and large).

V. INDICATIONS FOR USE

Cala kIQ is indicated to aid in the temporary relief of hand tremors in the treated hand following stimulation in adults with essential tremor.

Cala kIQ is indicated to aid in the temporary relief of postural and kinetic hand tremor symptoms that impact some activities of daily living in the treated hand following stimulation in adults with Parkinson's Disease.

VI. SUBSTANTIAL EQUIVALENCE

The Subject Device of this premarket notification is Cala kIQ Gen 2 with labeling updates, utilizing Cala kIQ Gen 1 as the predicate device. Cala kIQ Gen 1 (K222237) is legally marketed as a Class II device, subject to Special Controls as per 21 CFR 882.5897. Cala kIQ Gen 2 was released under note-to-file. Both Cala kIQ Gen 1 and Gen 2 are manufactured by Cala Health, Inc.

Cala kIQ Gen 1 and Gen 2 share:

- The same intended use
- The same control mechanisms, operating principles and energy type
- The same calibration algorithm and stimulation waveform
- The same architecture of Stimulator attached to Wristband with charging Base Station

The scope of this submission is only labeling changes to the currently-released Cala kIQ (Gen 2), including the following updates to the Patient and Healthcare Professional (HCP) Guides:

- Clarifying the contraindications language
- Updating warnings and cautions
- Adding troubleshooting steps for patient convenience in resolving band connection errors
- Simplifying patient instructions for accessing the Patient Portal

Table 1 below provides a summary comparison between the Subject Device, Cala kIQ Gen 2 with proposed labeling updates, and the Predicate Device, Cala kIQ Gen 1, with respect to intended use, technological characteristics, and principles of operation, providing detailed information regarding the basis for the determination of substantial equivalence.

Table 1. Cala kIQ Substantial Equivalence

	Predicate Device **Cala kIQ Gen 1** **K222237**	**Subject Device** **Cala kIQ Gen 2 with proposed labeling updates**	**Substantially Equivalent?**
510(k) Number	K222237	This submission	
Manufacturer	Cala Health	Cala Health	
Intended Use			
Intended Use	Delivery of transcutaneous afferent patterned stimulation ("TAPS") for treatment of hand tremors	Delivery of transcutaneous afferent patterned stimulation ("TAPS") for treatment of hand tremors	Yes Subject Device is identical to Predicate Device
Indications for Use	Cala kIQ is indicated to aid in the temporary relief of hand tremors in the treated hand following stimulation in adults with essential tremor. Cala kIQ is indicated to aid in the temporary relief of postural and kinetic hand tremor symptoms that impact some activities of daily living in the treated hand following stimulation in adults with Parkinson's disease.	Cala kIQ is indicated to aid in the temporary relief of hand tremors in the treated hand following stimulation in adults with essential tremor. Cala kIQ is indicated to aid in the temporary relief of postural and kinetic hand tremor symptoms that impact some activities of daily living in the treated hand following stimulation in adults with Parkinson's disease.	Yes Subject Device is identical to Predicate Device
Target Population	Adults with essential tremor or Parkinson's Disease	Adults with essential tremor or Parkinson's Disease	Yes Subject Device is identical to Predicate Device
Anatomical site	Median and radial nerves	Median and radial nerves	Yes Subject Device is identical to Predicate Device
Intended Users	Patient	Patient	Yes Subject Device is identical to Predicate Device

	Predicate Device Cala kIQ Gen 1 K222237	Subject Device Cala kIQ Gen 2 with proposed labeling updates	Substantially Equivalent?
Clinical Setting	In-home use after an initial calibration in the patient's home or in the physician's office.	In-home use after an initial calibration in the patient's home or in the physician's office.	Yes Subject Device is identical to Predicate Device
Rx or OTC use	Prescription Use only	Prescription Use only	Yes Subject Device is identical to Predicate Device
Design			
Technology	Transcutaneous Afferent Patterned Stimulation (TAPS) delivered through electrodes embedded on wearable band	Transcutaneous Afferent Patterned Stimulation (TAPS) delivered through electrodes embedded on wearable band	Yes Subject Device is identical to Predicate Device
Energy used or delivered	Electrical stimulation	Electrical stimulation	Yes Subject Device is identical to Predicate Device
Human Factors	Wrist-worn electrical stimulator with detachable band containing multi-use electrodes. Base station provides charging function and contains indicator lights for operating status	Wrist-worn electrical stimulator with detachable band containing multi-use electrodes. Base station provides charging function and contains indicator lights for operating status	Yes Subject Device is identical to Predicate Device
Patient Contacting Materials	The following components of the Cala kIQ Band have Surface, Intact Skin Long-Term contact (>30 days): • Elastic (polyester) • Electrodes (silicone with carbon nanotube dispersion) • Microfiber (polyurethane and nylon) • Elastic (nylon with spandex) • Band thread (polyester)	The following components of the Cala kIQ Band have Surface, Intact Skin Long-Term contact (>30 days): • Electrodes (silicone with carbon nanotube dispersion • Elastic (polyester with spandex) • Band thread (cotton) • Band silicone body • Band hook (sabic PC/ABS) • Stimulator bottom (PC/ABS)	Yes All patient-contacting materials are biocompatible per ISO-10993. The electrode material is unchanged from the predicate device.

	Predicate Device **Cala kIQ Gen 1** **K222237**	**Subject Device** **Cala kIQ Gen 2 with proposed labeling updates**	**Substantially Equivalent?**
Operating Principle	Transcutaneous Afferent Patterned Stimulation (TAPS) to the median and radial nerves of a patient's wrist	Transcutaneous Afferent Patterned Stimulation (TAPS) to the median and radial nerves of a patient's wrist	Yes Subject Device is identical to Predicate Device
Electrodes	6 Individual Electrode Surface area 22mm x 6mm = 1.32 cm^2	6 Individual Electrode Surface area 22mm x 6mm = 1.32 cm^2	Yes Subject Device is identical to Predicate Device
Battery and Base Station	Rechargeable Lithium-ion battery and AC-powered charger.	Rechargeable Lithium-ion battery and AC-powered charger.	Yes Subject Device is identical to Predicate Device
Performance			
User Workflow	1. Calibration 2. Set stimulation intensity 3. Therapy available on demand	1. Calibration 2. Set stimulation intensity 3. Therapy available on demand	Yes Subject Device is identical to Predicate Device
Treatment Time	40 minutes	40 minutes	Yes Subject Device is identical to Predicate Device
Wristband life	90 days	90 days	Yes Subject Device is identical to Predicate Device.

	Predicate Device **Cala kIQ Gen 1** **K222237**	**Subject Device** **Cala kIQ Gen 2 with proposed labeling updates**	**Substantially Equivalent?**
Shelf-life	Cala kIQ Band: 3 months	Cala kIQ Band: 18 months	Yes Increased shelf-life is due to availability of test data, not from changes in performance or safety characteristics
Output Specifications			
Waveform (e.g., pulsed monophasic, biphasic)	Biphasic	Biphasic	Yes Subject Device is identical to Predicate Device
Shape (e.g., rectangular, spike, rectified sinusoidal)	Rectangular	Rectangular	Yes Subject Device is identical to Predicate Device
Maximum Output Voltage (volts)	4 @ 500Ω	4 @ 500Ω	Yes Subject Device is identical to Predicate Device
	80 @ 10kΩ	80 @ 10kΩ	
Maximum Output Current (mA)	8 @ 500Ω	8 @ 500Ω	Yes Subject Device is identical to Predicate Device
	8 @ 10kΩ	8 @ 10kΩ	
Duration of primary (depolarizing) phase (µsec)	300	300	Yes Subject Device is identical to Predicate Device

	Predicate Device Cala kIQ Gen 1 K222237	Subject Device Cala kIQ Gen 2 with proposed labeling updates	Substantially Equivalent?
Pulse Duration (µsec)	650	650	Yes Subject Device is identical to Predicate Device
Frequency (Hz)	150	150	Yes Subject Device is identical to Predicate Device
Symmetrical phases?	Yes	Yes	Yes Subject Device is identical to Predicate Device
Phase Duration (µS)	300 each phase	300 each phase	Yes Subject Device is identical to Predicate Device
Net Charge (µC)	0 @500Ω	0 @500Ω	Yes Subject Device is identical to Predicate Device
Maximum Phase Charge (µC)	2.4 @ 500Ω	2.4 @ 500Ω	Yes Subject Device is identical to Predicate Device
Maximum Current Density (mA/cm^2, r.m.s.)	1.29 @ 500Ω	1.29 @ 500Ω	Yes Subject Device is identical to Predicate Device
Maximum Average Current (mA) (average absolute value)	0.72 @ 500Ω	0.72 @ 500Ω	Yes Subject Device is identical to Predicate Device

	Predicate Device Cala kIQ Gen 1 K222237	Subject Device Cala kIQ Gen 2 with proposed labeling updates	Substantially Equivalent?
Maximum Average Power Density (mW/cm²)	2.18 @ 500Ω (0.0022 W/cm^2)	2.18 @ 500Ω (0.0022 W/cm^2)	Yes Subject Device is identical to Predicate Device
Safety			
Electrical safety	Conforms to IEC 60601 Electrical Safety	Conforms to IEC 60601 Electrical Safety	Yes Subject Device is identical to Predicate Device
Compatibility with intended environments	Conforms to EMC requirements	Conforms to EMC requirements	Yes Subject Device is identical to Predicate Device
Mechanical safety	Conforms to IEC 60601 Electrical Safety	Conforms to IEC 60601 Electrical Safety	Yes Subject Device is identical to Predicate Device
Chemical safety	Not applicable	Not applicable	Yes Subject Device is identical to Predicate Device
Thermal safety	Conforms to IEC 60601 Electrical Safety	Conforms to IEC 60601 Electrical Safety	Yes Subject Device is identical to Predicate Device
Radiation safety	Not applicable	Not applicable	Yes Subject Device is identical to Predicate Device

VII. PERFORMANCE DATA

There was no performance testing conducted in support of Cala kIQ Gen 2 premarket application.

VIII. CLINICAL DATA

There was no clinical testing conducted in support of Cala kIQ Gen 2 premarket application.

IX. SUBSTANTIAL EQUIVALENCE

Substantial equivalence of the Subject Device, Cala kIQ Gen 2 with proposed labeling updates, to the Predicate Device, Cala kIQ Gen 1, is demonstrated based on the comparison of labeling, technical characteristics, performance, and testing.

The intended use and indications for use are unchanged.

The performance data reviewed in previous submissions, including the predicate submission, are still applicable to the subject device since they are similar in technology, and, hence, the previously reviewed data supports that the subject device meets special controls.

X. CONCLUSION

The Subject Device, Cala kIQ Gen 2 with labeling updates, has the same intended use and indications for use as the Predicate Device, Cala kIQ Gen 1. The subject and predicate device have similar technological characteristics and the differences do not raise new questions of safety or effectiveness.

Cala kIQ Gen 2 and Cala kIQ Gen 1 are substantially equivalent. Cala Health has provided evidence as described above to demonstrate that any differences in the technological characteristics do not impact safety or effectiveness.